The Barrier–Circumvention Contest:

Endogenous Networks, Path Dependence, and Hysteresis

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Abstract

The desire-lever operates within a network of principals and agents. But the network itself is contested. Incumbents invest in entry barriers; challengers invest in circumvention. This paper endogenizes the network: it models the barrier–circumvention contest as a game in which the effective barrier is a function of the incumbent's investment and the challenger's counter-investment, and it shows that the contest exhibits strategic complementarity. The paper derives the free-entry equilibrium, the endogenous-barrier equilibrium, and the conditions under which two pairwise stable networks coexist: a competitive one and a captured one. It proves that the captured equilibrium is an absorbing state, connected to the competitive equilibrium by path dependence and hysteresis. The barrier–circumvention contest is formally the same game as the discovery–concealment contest, at a different level of the theory: the discovery contest operates within a fixed network, while the barrier contest operates on the network itself.

Keywords: endogenous networks, entry barriers, circumvention, strategic complementarity, path dependence, hysteresis, competitive equilibrium, captured equilibrium

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1 The Network and Its Contest

The desire-lever operates within a network of principals and agents. When many principals hold levers, and none dominates, the agent is safer: the agent can play them against each other, and the agent's alternatives multiply. When the many collapse into one, the network becomes captured, and the extraction limit returns to the bilateral level.

But the network is not given. It is contested. Incumbents invest in entry barriers to keep competitors out; challengers invest in circumvention to get in. The contest over the network is the subject of this paper.

Remark 1.1 (The Structural Analogy)**.** *The barrier–circumvention contest and the discovery–concealment contest are formally the same game at different levels of the theory. The discovery contest operates* within *a fixed network, determining the terms of any link. The barrier contest operates* on *the network, determining whether the network is competitive or captured.*

2 The Free-Entry Equilibrium

Let N be the number of active principals, $\pi(N)$ the profit of an active principal, and η the fixed entry cost. The free-entry equilibrium satisfies

$$\pi(N^*) \geq \eta > \pi(N^* + 1). \tag{1}$$

Proposition 2.1 (Free Entry)**.** *The free-entry number of principals N^* is increasing in the value of the desired good v, decreasing in the entry cost η, and decreasing in the barrier investment E.*

The free-entry equilibrium is the competitive benchmark. When entry is cheap and barriers are low, many principals compete, and the agent's alternatives are many. When entry is expensive or barriers are high, few principals compete, and the agent's alternatives are few.

3 Endogenous Barriers

Incumbent principals can invest E in entry barriers. The incumbent's problem is

$$\max_{E \geq 0} \big(N^*(E) - 1\big)\, r\big(N^*(E)\big) - \psi(E), \tag{2}$$

where $r(N) = \pi(N) - \eta$ is the rent per incumbent and $\psi(E)$ is the cost of the barrier investment.

Proposition 3.1 (Endogenous Barriers)**.** *The optimal barrier investment E^* satisfies*

$$\frac{d}{dE}\Big[\big(N^*(E) - 1\big)\, r\big(N^*(E)\big)\Big] = \psi'(E^*). \tag{3}$$

E^* is increasing in v and decreasing in ψ'.

The incumbent invests in barriers up to the point where the marginal protected rent equals the marginal cost of the barrier. The higher the value of the desired good, the more the incumbent invests; the more expensive the barrier, the less.

4 Multiplicity and Coordination Failure

Proposition 4.1 (Multiplicity)**.** *If the barrier cost ψ is high and the entry cost η is low, there are two pairwise stable networks: a* competitive *one with N^* large and $E^* = 0$, and a* captured *one with N^* small and $E^* > 0$.*

Sketch. In the competitive equilibrium, no incumbent invests in barriers because the cost exceeds the protected rent. In the captured equilibrium, incumbents invest because the protected rent is large enough. Both satisfy pairwise stability. □

The multiplicity is a coordination failure. The network can be competitive or captured, and both are stable. Which one obtains depends not on the parameters but on the history: the network that is competitive stays competitive, and the network that is captured stays captured.

5 Path Dependence and Hysteresis

Proposition 5.1 (Hysteresis)**.** *Starting from a captured network, the equilibrium path satisfies $dE_t/dt > 0$ and $d\kappa_t/dt > 0$, where κ_t is the degree of capture. The captured equilibrium is an absorbing state.*

The captured equilibrium is an absorbing state. Once the network is captured, it stays captured, because the incumbents' barrier investment raises the cost of entry, which reduces the number of principals, which raises the protected rent, which justifies more barrier investment. The captured equilibrium is self-reinforcing.

The competitive equilibrium is not an absorbing state in the same way. A competitive network can be captured by a shock—a merger, a regulatory change, a technological shift—that tips it into the captured equilibrium. But a captured network cannot be un-captured by a symmetric shock, because the barriers are already in place.

This is hysteresis: the network's state depends on its history, not just on its current parameters. The same parameters can support either equilibrium, and the equilibrium that obtains is the one that was reached first.

6 The Barrier–Circumvention Contest

The barrier technology is contested. Incumbents invest E in barriers; challengers invest X in circumvention. The *effective* barrier is

$$B = B(E, X), \quad B_E > 0, \quad B_X < 0, \quad B_{EE} < 0,\ B_{XX} > 0. \tag{4}$$

The challenger's problem is

$$\max_{X \geq 0}\ \rho\big(B(E,X)\big)\, r\big(N^*(B)\big) - \xi X, \tag{5}$$

and the incumbent's problem is

$$\max_{E \geq 0}\ \big[1 - \rho\big(B(E,X)\big)\big]\,\big(N^*(B) - 1\big)\, r\big(N^*(B)\big) - \psi(E). \tag{6}$$

Proposition 6.1 (Barrier–Circumvention Contest)**.** *In the simultaneous-move game, the equilibrium* (E^*, X^*) *satisfies the two first-order conditions. The contest exhibits strategic complementarity:* $\partial E^*/\partial X^* > 0$ *and* $\partial X^*/\partial E^* > 0$*.*

Proof. Differentiate the first-order conditions with respect to the opponent's choice. The cross-partial $\partial^2 \Pi^{inc}/\partial E\, \partial X = -\rho''\, B_E\, B_X\, (N^* - 1)r > 0$ and $\partial^2 \Pi^{ch}/\partial X\, \partial E = -\rho''\, B_X\, B_E\, r > 0$. Both are positive, giving best-response monotonicity. □

The strategic complementarity means that the contest escalates. The incumbent's barrier investment raises the challenger's circumvention investment, and the challenger's circumvention investment raises the incumbent's barrier investment. The contest is an arms race.

Corollary 6.1 (Barrier Contest and Longevity)**.** *In the long run, the captured equilibrium is sustainable only if the incumbent maintains a positive effective barrier* $B^* > 0$ *in the stationary state. If the challenger's circumvention technology is sufficiently cheap, the stationary* B^* *converges to zero, and the captured equilibrium unravels.*

7 Connection to the Win–Win Equilibrium

Proposition 7.1 (Win–Win in the Network)**.** *The win–win equilibrium exists in the network if and only if the network is in the competitive equilibrium and the alignment condition holds. In the captured equilibrium, the win–win equilibrium does not exist.*

Proof. In the competitive network, the agent's outside option is high, so the extraction limit is low, and the win–win equilibrium exists. In the captured equilibrium, the agent's outside option is destroyed, so the extraction limit is restored to the bilateral level, and the win–win equilibrium does not exist. □

This is the deepest connection between the network and the ethics of the lever. The win–win equilibrium—the region in which the principal restrains extraction because restraint is durable—exists only in the competitive network. In the captured network, the principal has no incentive to restrain, because the agent cannot leave. The ethics of the lever are a property of the network, not of the principal.

8 Conclusion

The desire-lever operates within a network, and the network is contested. Incumbents invest in barriers; challengers invest in circumvention. The contest exhibits strategic complementarity, and it produces two pairwise stable equilibria: a competitive one and a captured one.

The captured equilibrium is an absorbing state, connected to the competitive equilibrium by path dependence and hysteresis. Once the network is captured, it stays captured, because the barriers are self-reinforcing.

The barrier–circumvention contest is formally the same game as the discovery–concealment contest, at a different level of the theory. The discovery contest operates within a fixed network; the barrier contest operates on the network itself.

The win–win equilibrium exists only in the competitive network. In the captured network, the principal has no incentive to restrain, because the agent cannot leave. The ethics of the lever are a property of the network, not of the principal.

The question is always the same: whose ledger is this, and for whom? For the captured network, the answer is: the incumbents', and at the challengers' and the agents' expense. The correction is to keep the network competitive—to keep the barriers low, the circumvention cheap, and the many from collapsing into one.

The rest is history—or its absence.